

LPU Learners Hub

Ubuntu Deployment Guide

Step-by-step instructions to deploy the Student Hub on an Ubuntu server using Git, Docker, and systemd.

Repository: <https://github.com/norielgecolea/lpu-student-hub.git>

Domain: leanershub.lpulaguna.com

Note: This project is a static Angular app. Add Dockerfile, nginx.conf, and docker-compose.yml to the repo root before step 5 if they are not present yet.

1. Install Git

Update packages and install Git:

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y git
git --version
```

2. Install Docker

Install Docker Engine and the Docker Compose plugin:

```
sudo apt install -y ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
  sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

echo "deb [arch=$(dpkg --print-architecture) \
  signed-by=/etc/apt/keyrings/docker.gpg] \
  https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io \
  docker-buildx-plugin docker-compose-plugin

sudo usermod -aG docker $USER
newgrp docker

sudo systemctl enable --now docker
docker --version
docker compose version
```

Log out and back in if docker commands still require sudo after newgrp.

3. Clone the Repository

Create an app directory and clone the project:

```
sudo mkdir -p /opt/apps
sudo chown $USER:$USER /opt/apps
cd /opt/apps

git clone https://github.com/norielgecolea/lpu-student-hub.git
cd lpu-student-hub
```

For a private repo, use SSH or a personal access token with HTTPS.

Required files in the project root

Dockerfile:

```
FROM node:22-alpine AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build:gh-pages

FROM nginx:1.27-alpine
COPY nginx.conf /etc/nginx/conf.d/default.conf
COPY --from=build /app/dist/student-hub/browser /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

nginx.conf:

```
server {
    listen 80;
    server_name _;
    root /usr/share/nginx/html;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

docker-compose.yml:

```
services:
  student-hub:
    build: .
    container_name: student-hub
    restart: unless-stopped
    ports:
      - "${HOST_PORT:-8080}:80"
    env_file:
      - .env
```

4. Create the .env File

Create environment settings used by Docker Compose:

```
cd /opt/apps/lpu-student-hub
nano .env
```

Example contents:

```
HOST_PORT=8080
APP_NAME=lpu-student-hub
APP_ENV=production
DOMAIN=leanershub.lpulaguna.com
```

```
chmod 600 .env
```

Keep .env out of git. It mainly controls the published host port for this static app.

5. Start Docker Compose as a Service

Create a systemd unit so the app starts on boot:

```
sudo nano /etc/systemd/system/student-hub.service
```

Service file contents:

```
[Unit]
Description=LPU Student Hub (Docker Compose)
Requires=docker.service
After=docker.service network-online.target
Wants=network-online.target

[Service]
Type=oneshot
RemainAfterExit=yes
WorkingDirectory=/opt/apps/lpu-student-hub
ExecStart=/usr/bin/docker compose up -d --build
ExecStop=/usr/bin/docker compose down
TimeoutStartSec=0

[Install]
WantedBy=multi-user.target
```

Enable and start the service:

```
sudo systemctl daemon-reload
sudo systemctl enable --now student-hub.service
sudo systemctl status student-hub.service
```

Verify the app is running:

```
docker compose ps
curl -I http://127.0.0.1:8080
```

After code updates:

```
cd /opt/apps/lpu-student-hub
git pull
sudo systemctl restart student-hub.service
```

Quick checklist

1. Install Git
2. Install Docker + Compose plugin
3. Clone the repository
4. Create .env
5. Enable student-hub.service and confirm http://SERVER:8080